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Article in *Fundamental and Applied Limnology / Archiv für Hydrobiologie* · January 2007

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Biological traits of benthic macroinvertebrates in California mediterranean-climate streams: long-term annual variability and trait diversity patterns

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With 6 figures, 6 tables and 2 appendices

Abstract: We examined long-term variability of the biological traits of benthic macroinvertebrate communities using data collected annually (20-, 7-, and 8-years in duration, respectively) from three study regions in northern California's mediterranean-climate region. We related temporal variability to several climatic and physical variables representing precipitation and hydrologic conditions, and assessed the stability of biological-trait composition using time-lag regression analysis. We also examined the temporal variability of trait diversity metrics and their relationship to taxa richness. Trait modalities exhibited high temporal variability (~ 20 % of all trait modalities had a CV > 50 %) and biological-trait composition varied along a gradient from dry (extreme drought) to wet (above average) years. Traits conferring resistance or resilience to drying (e.g. desiccation resistance, aerial respiration) were more common in drought years, whereas traits conferring resistance or resilience to floods (e.g. flattened body shape, drift dispersal) were more common in wet years. Furthermore, prolonged drought and the patterns of increasing or decreasing precipitation over time in the study regions resulted in patterns of directional change in biological-trait composition. Although biological traits had high temporal variation within each region, between-region variation was higher than temporal variation in a single region. As found in other studies, trait diversity and richness metrics were relatively invariable over time despite high taxonomic turnover. However, there was considerable regional variation in trait diversity metrics along a gradient of increasing taxonomic diversity. The four functional metrics studied showed high redundancy. Taxa richness was positively correlated with three functional metrics (trait richness and two trait diversity measures), and trait richness and diversity became saturated taxa richness increased (~50 taxa). In communities largely controlled by abiotic factors, such as stream invertebrates in mediterranean-climate, high redundancy may be the result of strong filtering of traits at larger spatial scales.

Key words: drought, long-term data, persistence, stability, streams, temporal variation.

Introduction

Variability in ecological communities arises from both spatial and temporal gradients in biotic interactions, and from abiotic factors that control the abundances of populations. In stream ecosystems, spatial (e.g. Townsend et al. 1997, Daufresne et al. 2003) and

some aspects of short-term temporal variation, such as seasonality (e.g. Snook & Milner 2002, Bêche et al. 2006), have been widely recognized and studied. In contrast, the assessment of the temporal variability of stream communities over multiple generations, which requires multi-year datasets (often longer than 5–10 years), is rare. The lack of published long-term

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studies limits both our understanding of, and perhaps leads to an underestimation of, natural variability because short-term studies generally fail to capture infrequent events (e.g. 100-year floods, fires) or long-term cycles in climate (e.g. multi-year droughts) (e.g. Eby et al. 2003). Furthermore, in assessing the impacts of management practices or other anthropogenic effects, long-term data may also be required to accurately differentiate the effects of natural variability (e.g. as a result of flooding or drought) from the impacts of interest, particularly in regions where high natural variability in the physical environment exists (such as in the mediterranean-climate, Gasith & Resh 1999).

Biological traits, e.g. locomotion, body shape, and life span, have the potential to elucidate patterns of temporal variability of stream communities because they can provide a mechanistic understanding of natural variation (e.g. Resh et al. 1994, Townsend et al. 1997). Furthermore, most studies that have examined the relationship between spatial- or temporal-environmental variation and community dynamics usually invoke biological traits in the discussion of their results (e.g. Boulton et al. 1992). This approach is logical because the explanation of why certain taxa respond in a predictable fashion to environmental change is usually a function of their life-history, behavior, morphology, or other adaptations (Lytle & Poff 2004).

The amount of temporal and spatial variation in biological traits has also been used to assess the diversity, stability, and redundancy of communities across multiple scales (e.g. Petchey & Gaston 2002). Benthic macroinvertebrates account for much of the biodiversity in streams, and they have important and often dominant roles in ecosystem function. Therefore, an understanding of the factors influencing the presence and relative abundance of their traits over time can provide insight into the functional diversity of the community (e.g. Heino 2005). For example, although it is widely assumed that functional diversity is strongly controlled by taxa diversity or richness, there is little information on the relationships between these two aspects of communities in most systems (e.g. Naeem 1998). Furthermore, because taxa richness and diversity in stream systems are sensitive to variability in disturbance-causing agents (e.g. floods, droughts), the extent of functional redundancy in stream systems has important implications for the temporal variability of ecosystem processes, as well as conservation of ecosystem function (e.g. Schwartz et al. 2000).

High functional redundancy among stream macroinvertebrates may be a contributing factor to recent studies that have reported relatively stable biological

trait composition over broad spatial scales and between seasons (Statzner et al. 2004, Bêche et al. 2006). The reason behind high functional redundancy in streams is not well understood, but it may arise from strong abiotic pressures imposed by the stream environment, which limits the number and combination of traits that can successfully result (e.g. Poff 1997). However, the data used in spatial studies of trait stability have been collected sporadically over several decades, from many sources, over a wide geographical region (e.g. all of western Europe), collected for a multitude of purposes, by different operators, and based on various sampling and identification methods (Statzner et al. 2004). Furthermore, the stability observed in these European temperate streams may be a result of relatively stable climate and precipitation on an annual scale relative to that in more variable areas such as in the mediterranean-climate region. A consistently collected dataset from the same sites, representing a long time-period and encompassing a range of abiotic conditions is essential to thoroughly assess the temporal variability of biological trait composition and aspects of functional diversity.

In this study, we examined 19–20 years of macroinvertebrate collections from four sites in two northern California streams to determine: (1) the influence of physical habitat and climatic factors on the temporal variability of biological traits; (2) the long-term stability of trait composition; and (3) the temporal variability of composite metrics of traits and their relationship to taxonomic diversity and richness. To determine the generality of these findings, we compared trends observed with those from collections (7–8 year duration) from two other northern California regions consisting of six additional sites.

Methods

Site descriptions

Hunting and Knoxville Creeks (20-year dataset)

Benthic macroinvertebrates were collected at four sites in Hunting and Knoxville Creeks (HKC), located approximately 30-km north of Lake Berryessa in the University of California McLaughlin Nature Reserve (Napa and Lake Co.), California. Collections were made from 1984–2003 at three sites (1P, 2P, 2D) in Hunting Creek, and one site in the adjacent Knoxville Creek (1D; Table 1). Once every year, 5 Surber samples (0.093 m², 0.5 mm mesh) were collected in 5 randomly selected riffles (the same riffles were sampled each year). Samples were collected on 15 April (± 3 d) in the post wet-season. Detailed descriptions of the study sites are given by Resh et al. (2005). Study site names indicate stream order (1 or 2) and permanence

Table 1. Summary of physical characteristics of the study sites.

	Latitude (N)	Longitude (W)	Elevation (m)	Stream order	Watershed area (km ²)	Flow regime	Total precipitation Mean $\pm$ SD (CV)	Range of total precipitation (mm) and years
Hunting/Knoxville Creeks (HKC)							605 $\pm$ 232 (38.4)	289 – 1062 (1990) (1995)
1D	38°47'56"	122°18'53"	390	1	2.1	ephemeral		
1P	38°51'56"	122°24'54"	634	1	4.4	intermittent		
2D	38°48'30"	122°22'36"	348	2	29.3	ephemeral ¹		
2P	38°49'45"	122°22'45"	402	2	22.1	perennial		
Big Sulphur Creek (BSC)							955 $\pm$ 457 (50.6)	238 – 1518 (1977) (1983)
BSC	38°48'3"	122°48'36"	415	3	35.0	perennial		
Blodgett Forest Creeks (BFRS)							1680 $\pm$ 619 (36.8)	664 – 2598 (2002) (1995)
BC	38°54'44"	120°39'25"	1380	1	1.0	perennial		
DA	38°54'21"	120°38'48"	1452	1	1.3	perennial		
DE	38°55'13"	120°40'14"	1280	1	1.4	perennial		
GC	38°52'12"	120°38'31"	1243	2	4.9	perennial		
MC	38°54'17"	120°38'39"	1310	1	2.4	perennial		

¹ The channel from which samples are taken is ephemeral, but it is adjacent to a perennial channel.

during the dry season (P = perennial, D = dry). For example, site 1P is a 1st-order perennial site (Table 1).

Arguably, the HKC dataset is the most consistently collected and analyzed long-term information available for benthic macroinvertebrates. The same persons collected, sorted, and identified all the samples over the 20-year period of the study. Invertebrates were usually identified to genus, with species distinguished as operational taxonomic units, (OTUs sensu Resh 1979); however, some non-chironomid Diptera and non-insects were identified to higher levels.

Big Sulphur Creek (7-year dataset)

Benthic macroinvertebrates were collected at one site on Big Sulphur Creek (BSC) from 1977–1983. BSC is a 3rd order perennial stream located in the California Coast Range in north-eastern Sonoma Co., California. A detailed description of this stream and site is given by McElravy et al. (1989). Once per year in the post wet-season (15 May $\pm$ 5 d), 4 Surber samples (0.093 m², 0.5 mm mesh) were collected in a random design, stratified to riffle and pool areas (with the same areas being sampled each year). Riffle and pool samples were kept separate for identification and analysis.

Blodgett Forest Creeks (8-year dataset)

Benthic macroinvertebrates were collected annually from perennial streams located in mixed-conifer forest in the central Sierra Nevada mountains at Blodgett Forest Research Station (BFRS) in El Dorado Co., California (Table 1). Detailed descriptions of these study sites are given by Bêche et al. (2005) and Resh et al. (2005). The 5 study sites were located in four 1st order streams in Bacon (BC), Dark Canyon (DC), Deep Canyon (DE), and Mutton Creeks (MC), and one 2nd order stream (Gaddis Creek, GC). Each year, 5 D-frame samples (1 m², 1 mm mesh) were taken in riffle and run habitats (in the same 50 m reach) at each

site. Post dry-season collections were made every 1 October ($\pm$ 3 d) from 1995–1998 and 2000–2003.

Climatic patterns: precipitation

The study areas are in California's mediterranean-type climate region, which is characterized by cool, wet winters and hot, dry summers, with 75 to 95 % of the precipitation occurring from October to May. Although the seasonal pattern of precipitation in the mediterranean climate is predictable, between-year totals of precipitation are extremely variable.

The following parameters were calculated based on precipitation data: total precipitation (October 1 to sample date) for all sites, the precipitation occurring one month prior to sampling (HKC and BSC only), and precipitation two weeks prior to sampling (HKC only). The 1-month and 2-week precipitation metrics were calculated to reflect the timing of precipitation (and floods) prior to the sampling event. However, only monthly data were available for BSC and, at BFRS there was no precipitation during the several months prior to October sampling.

Physical habitat: shear stress and flow

FST (Fließwasserstammtisch)-hemisphere values (Statzner & Müller 1989) were used to measure hydraulic forces (shear stress) at the stream bottom where samples were taken at each HKC site from 1989 to 1998; approximately 10 measurements were taken per sampling date and location. The mean FST-hemisphere value for each site-year was used in subsequent analyses. Each hemisphere measurement provides an estimate of the minimum, bottom shear-stress force in dyn cm⁻¹ required to make it move (e.g. hemisphere 0 corresponds to 0.8 dyn cm⁻¹ and hemisphere 16 corresponds to 44.8 dyn cm⁻¹; Statzner et al. 1991). Furthermore, discharge measured at the time of sampling was used to for all correlation analyses.

Table 2. Biological traits (15) and modalities (66) used in this study. In figures and tables, most traits are referred to by the letter code (e.g., a1).

Maximum size (mm)		Parental care		Body armoring	
a1	< 5	g1	isolated eggs	l1	none (soft-bodied)
a2	5–10	g2	clutches	l2	moderate (sclerotized)
a3	10–20	g3	terrestrial	l3	strong (snails/stone cased)
a4	20–40	g4	endophytic		
a5	> 40	g5	ovoviparous		
		g6	budding	Feeding habits	
Voltinism				m1	absorber
b1	bi- to multivoltine	Dispersal		m2	deposit feeder
b2	univoltine	h1	aquatic passive	m3	shredder
b3	semi- to merovoltine	h2	aquatic active	m4	scraper
		h3	aerial passive	m5	filterer
Adult longevity		h4	aerial active	m6	piercer
c1	≤ 1 d			m7	engulfer
c2	> 1–10 d	Attachment (aquatic stage)		m8	parasite
c3	> 10–30 d	i1	swimmer		
c4	> 30d to 365 d	i2	crawler	Food eaten	
c5	> 365 d	i3	burrower (epibenthic)	n1	FPOM < 1 mm
		i4	endobenthic (interstitial)	n2	CPOM > 1mm
Aquatic stage		i5	temporarily attached	n3	periphyton
d1	egg	i6	permanently attached	n4	macrophyte (live)
d2	larvae			n5	dead animals > 1 mm
d3	adult	Streamlining		n6	animals ≤ 1mm
		j1	streamlined	n7	animals > 1 – 10 mm
Resistance		j2	not streamlined	n8	animals > 10 mm
e1	desiccation				
e2	diapause	Body shape		Respiration	
e3	none	k1	flattened	o1	cutaneous
		k2	cylindrical	o2	gills
Reproduction		k3	spherical	o3	plastron
f1	sexual			o4	aerial
f2	asexual				
f3	parthenogenic				

Flow was measured at the time of sampling at each site by measuring channel characteristics (width, depth) and velocity. Flow data from gauges (DC, DE, GC), models (HKC sites) or a combination of both (BSC, BC, MC) were used in conjunction with sediment characteristics to determine the number of times the bed moved (equal to the movement of the median particle size, D_{50}) during the wet season.

Biological traits

We used 16 biological traits related to body morphology (size, body shape, body armoring), life history (voltinism, reproduction, parental care, adult longevity, resistance), attachment, respiration, dispersal, and feeding (habits and food eaten) of the 263 taxa collected (Table 2). These traits were selected because they may be expected to vary interannually in response to changes in precipitation or climate. The 16 biological traits were divided into 66 modalities (e.g. the trait ‘maximum body size’ has 5 modalities, ranging from < 5 to > 40 mm). Trait information was collected from the literature (frequently used sources among the > 300 citations include: Usinger 1956, Mer-

ritt & Cummins 1996, Wiggins 1996, Stewart & Stark 2002, among others); a complete list of literature is available by request. Trait information was coded at the generic level, except for some Diptera and non-insect taxa, which were coded at the family or order level, respectively. Each taxon is given a score (between 0 and 3, or 0 and 5), corresponding to its affinity to the modality using fuzzy coding (Chevenet et al. 1994). Trait information was available for all benthic macroinvertebrate taxa collected from all sites. Traits were rescaled as proportions (sum = 1), such that for each trait modality, values ranged from 0 (no affinity among individuals for the modality) to 1 (all individuals had exclusive affinity for the modality).

Data analysis

Temporal trends in biological trait composition

To examine overall differences between the sites, we calculated the mean, standard deviation (SD), and coefficient of variation (CV as SD/mean) for each trait modality at each site and for each year separately. To examine patterns between these met-

Table 3. Description of taxa and trait diversity metrics calculated in this study.

Data type	Data format	Diversity measure	Abbreviation	Formula	Rationale	Citation/Source
Taxa	n/a	Taxa richness	Taxa richness	number of taxa present		
Taxa	raw	Simpson's Diversity	Taxa diversity	$S = 1 - D, \text{ where } D = \sum \left(\frac{n_i}{N} \right)^2$ $n_i = \text{abundance of species } i$ $N = \text{total abundance of all species}$	Probability that two individuals randomly selected from a sample will belong to different species	
Traits	n/a	Trait richness	TR	number of trait categories represented		e.g. Heino (2005)
Traits	log10-weighted traits	Trait diversity	TD	Average S across all traits where $S_i = 1 - D_i, \text{ where } D_i = \sum \left(\frac{n_j}{N} \right)^2$ $D_i = \text{Simp. diversity of trait } i$ $n_j = \text{rel. abundance of trait category } j$ $N = \text{total abundance of all traits categories}$	Describes the number of trait categories per trait and the categories within a trait. Taken as an average of S for each trait to reduce lack of independence among traits.	e.g. Heino (2005)
Traits	n/a	Functional diversity	FD	$FD = \text{pairwise dist. between taxa}$ where distance was calculated using Bray-Curtis	Analogous to using the total branch length from a cluster analysis of taxa x traits matrix	Petchey & Gaston (2002)
Traits	log10-weighted traits	Rao's quadratic entropy	RD	$RD = \sum_{i=1}^s \sum_{j=1}^s d_{ij} p_i p_j$ where d_{ij} is the Bray-Curtis distance between the i th and j th species	Probability that two individuals randomly selected from a sample will have the same trait profile (if trait profiles are the same, index is equivalent to Simpson's Diversity)	Rao (1982), Ricotta (2005), Zoltán (2005)

rics and physical variables, we calculated the non-parametric Spearman rank correlation between each trait modality, and each physical and climatic variable for each site separately.

Multivariate ordination was used to explore temporal patterns in biological trait composition. Separate ordinations were done for wet-season samples for each of the regions (HKC, BSC, BFRS) by combining sites within an area. Abundance-weighted biological traits were analyzed using fuzzy correspondence analysis (FCA, Chevenet et al. 1994) using the *ade4* package (Thioulouse et al. 2004) in R 2.2.1 (R Development Core Team 2005), resulting in four FCA ordinations (i.e. all regions combined, HKC, BSC, BFRS). FCA is an eigen-based ordination method that is an application of multiple correspondence analysis to fuzzy-coded data designed to analyze data that are in multiple categories with subcategories (Chevenet et al. 1994). Spearman rank correlation was used to assess the relationship between each of the measured physical/climatic variables and year of sampling with each of the ordination axes.

Long-term trends in variability of biological trait composition

To examine whether directional change had occurred during the study period, we used time-lag regression analysis (TLR, Collins et al. 2000). TLR examines ecological dissimilarity and time lags (1 to 19 years) between all pairs of observations. The Bray-Curtis distance was calculated for all pairs of sample dates for each site based on abundance-weighted traits. Bray-Curtis distance was then regressed against the square root of the time lag between those samples. The square root transformation reduces the potential effect of the few data points at long time-lags on the regression (Collins et al. 2000). Because higher distance values indicate communities that are more dissimilar, this analysis allows for determination of the type of temporal change occurring: a positive relationship indicates directional change, a negative relationship indicates convergence on a previous community structure, and no relationship ($r^2 \approx 0$) indicates either stability or stochastic change (Collins et al. 2000). A Mantel test between distance in time (time lag) and ecological distance with 10,000 Monte-Carlo simulations was used to determine whether the correlations obtained were significant using PC-ORD (v. 4.27, McCune & Mefford 1999).

Temporal variability of trait diversity and richness

The diversity and richness of traits represented at a given site and year was estimated using four methods: (1) trait richness (TR); (2) trait diversity (TD), which was calculated using Simpson's (1-D) diversity metric and was based on abundance-weighted traits; (3) functional diversity (FD) which was calculated using the Bray-Curtis distance metric and was based on presence/absence data; and (4) Rao's diversity coefficient (RD; Rao 1982, Ricotta 2005) which was calculated using both species differences based on their traits and the relative abundance of those species (Table 3). TR was determined as the number of trait categories represented in a given site and year. TD was determined by calculating Simpson's diversity across all modalities within each trait and then averaged across all 16 traits for each site-year using $\log_{10}$ -abundance weighted trait values. FD was calculated by taking the average Bray-Curtis distance between all taxa present in a given site-year (based on their traits), and represents the average distance between any two taxa in the sample. RD was calculated by first obtaining a distance ma-

trix of species based on their traits and then using the relative abundance of species in the following equation to describe the probability that two randomly selected individuals will have the same trait profile (Table 3).

Temporal patterns in diversity and richness of traits were examined by examining deviations from long-term means. The variability of trait diversity and richness was determined by examining the deviation of each parameter from the long-term average. The relationship between TD, FD, and TR and taxa diversity (Simpson's 1-D) and richness were examined for all sites across years. Furthermore, relationships between trait metrics and physical and climatic variables were examined using Spearman rank correlations.

Results

Climatic patterns: precipitation

Wet-season precipitation varied considerably during the study periods and significant temporal trends in total precipitation were identified in each region (Fig. 1). For example, at HKC, total precipitation increased over the years studied (Fig. 1), which was a result of the dry conditions that prevailed in the first decade (1984–1994), and the wet conditions that occurred in the latter decade (1995–2003), of the study. Furthermore, the study period encompassed a six-year period of extended drought (1987–1992), when rainfall was within or close to the driest quartile based on a 66-year record. Although total precipitation was not significantly correlated with year at BSC ($r_s = 0.54$, $p = 0.22$; Fig. 1), 1977 was the driest year on record for the previous 108 years. At BFRS, there was a significant decrease in precipitation during the study period ($r_s = -0.86$, $p = 0.007$; Fig. 1), and 2002 was the driest year in the 31-year record. Furthermore, there were distinct differences in precipitation between the two sample periods at BFRS, with the sample period from 1995 to 1998 being very wet (rainfall each year > 1 SD above the mean, except 1996) and the sample period from 2000 to 2003 being average (2000) to very dry (2001–2003, > 1 SD below the mean).

Physical habitat: shear stress and flow

Shear stress at the time of sampling varied among sites, as indicated by significant differences of the average FST-hemisphere value among sites (ANOVA, $df = 3$, $F = 26.6$, $p < 0.001$). Like rainfall, shear stress increased over time at all of the sites, and this relationship was strongest at site 1P. Shear stress was also positively correlated with total precipitation at sites 1P ($r_s = 0.76$, $p = 0.01$) and 2D ($r_s = 0.79$, $p = 0.006$) and was strongly correlated with 2-week precipitation at

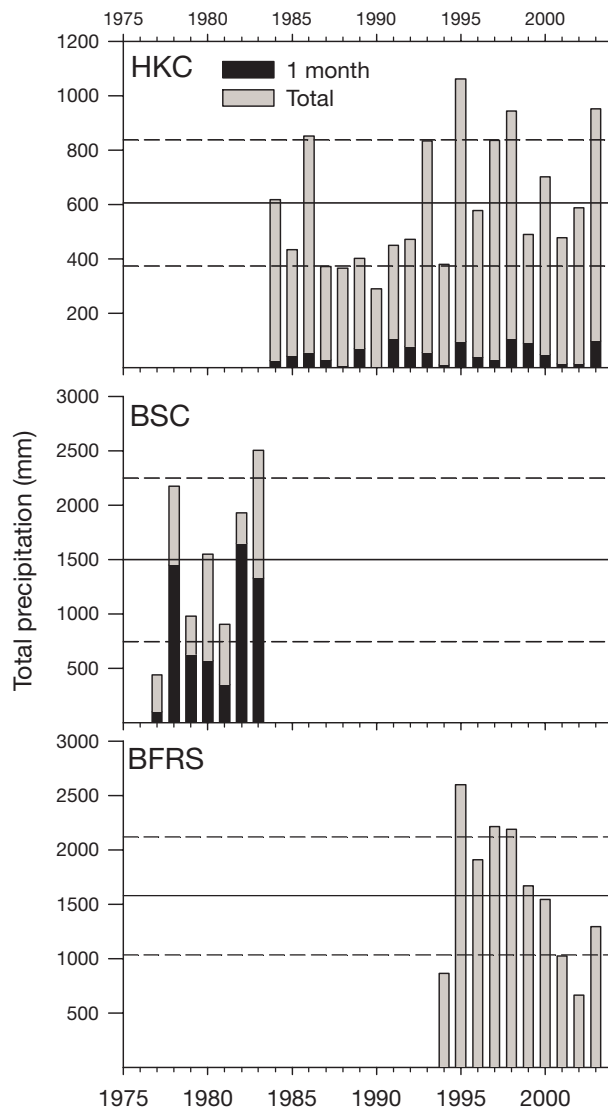


Fig. 1. Total wet season (October to sample date) precipitation (in mm, gray bars) and precipitation 1-month prior to sampling (black bars) at each study region. Precipitation 1-month prior to sampling was always zero at BFRS. The solid and dotted lines represent the long-term mean precipitation $\pm$ standard deviation, respectively.

all HKC sites ($r_s > 0.76$, $p < 0.01$) except site 2P ($r_s = 0.50$, $p = 0.14$), where 1-month precipitation was a better predictor ($r_s = 0.64$, $p = 0.05$).

Temporal trends in biological trait composition

Most traits exhibited low variability over the 7 to 20 year study period, with the average trait CV ranging from 24.7 % at site D2 (BFRS) to 37.5 % at BSC pool (Appendix 1). Traits exhibiting the highest variability were those that were represented by very few taxa that

occurred in low abundance in only a few years, whereas the lowest variability was found among those traits that represented the majority of all abundant taxa. For example, “sexual reproduction” (f1) was the least variable among all of the trait categories, with CV ranging from 0.86 % to 1.62 %. Similarly, “laying clutches of eggs” (g2) was also relatively invariable, with CV ranging from 1.8 % to 6.3 % (Appendix 1). The most variable traits included “parthenogenesis” (f3; CV ranging from 55.7 % to 189.4 %), “permanent attachment” (i6; CV ranging from 50.2 % to 282.8 %), “budding” (g6; CV ranging from 135.3 % to 447.2 %), “piercing feeding habits” (m6; CV ranging from 45.0 % to 136.7 %), and “feeding on animals > 10 mm” (n8; CV ranging from 89.8 % to 198.4 %) (Appendix 1). These highly variable traits were represented by few taxa, and only “piercing feeding habits” (m6) was present ≥ 1 year during the study period at all sites.

Multivariate patterns in biological trait composition

All regions. The FCA of all three regions showed that trait composition differed most between the two regions having the most divergent climates, HKC and BFRS, with BSC having an intermediate trait composition and climate (Fig. 2). The FCA resulted in a 2-axis solution representing 56.5 % of the total inertia of 0.06. Axis 1 represented 40.0 % of the total inertia and axis 2 represented 16.5 %. In addition to site differ-

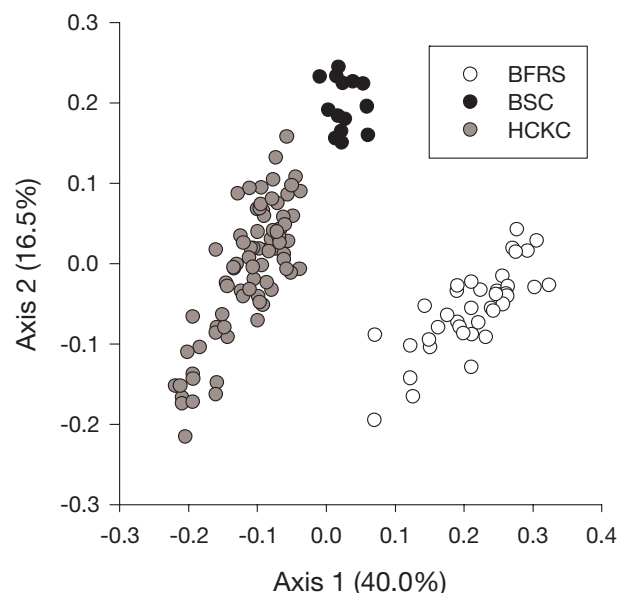


Fig. 2. FCA ordination of biological traits in site-year space for all regions combined. Site-years are represented by gray circles for HKC, black circles for BSC, and open circles for BFRS. 56.5 % of the variance was explained in 2 axes.

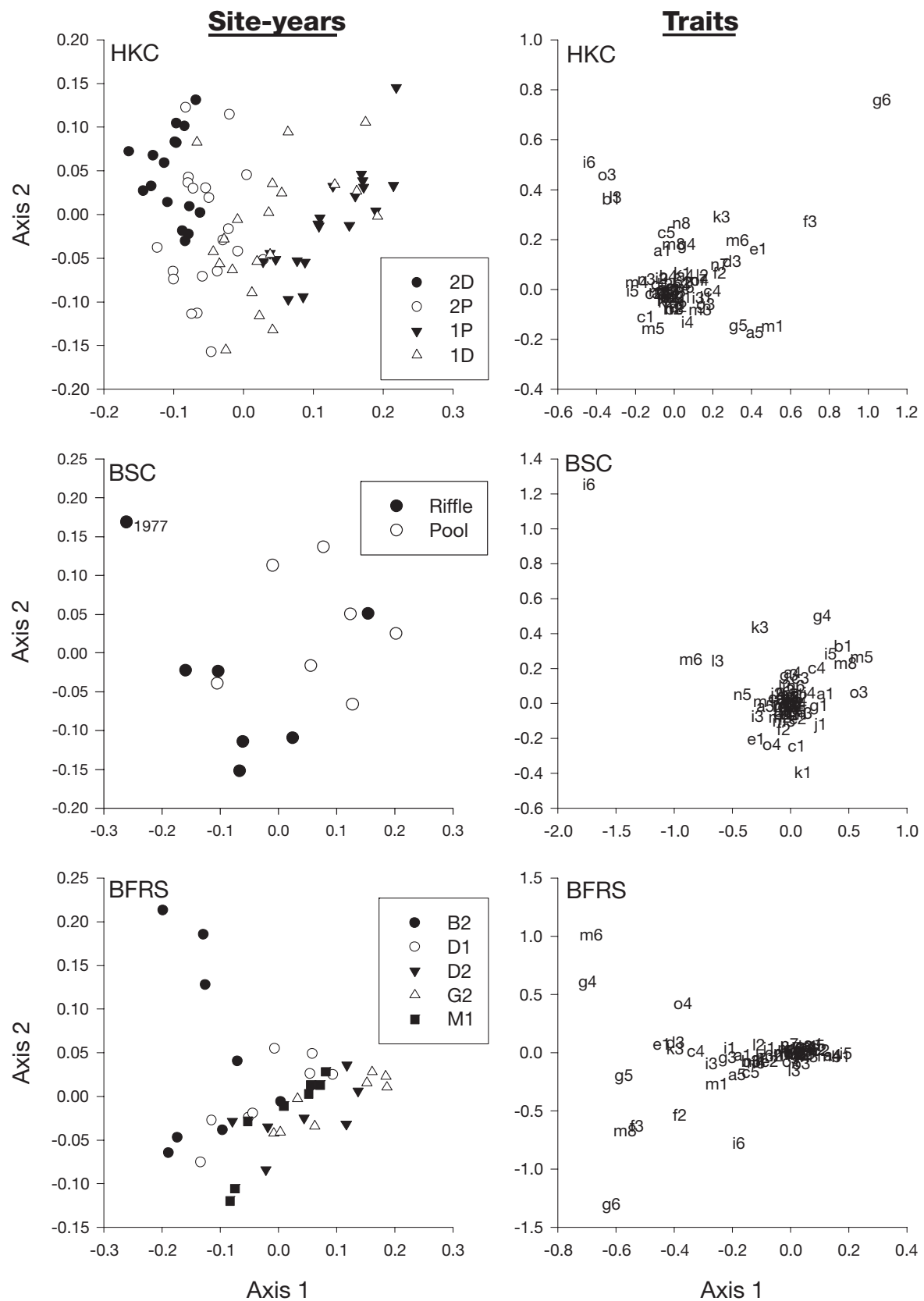


Fig. 3. FCA ordinations of biological traits in site-year space (left column) and traits-space (right column) for each study region. In all three ordinations, a 3-axis solution was found; axis 3 is not shown. Sites or habitats sampled within each region are indicated by different symbols. Trait codes are listed in Table 2.

Table 4. Spearman rank correlation (r_s) between each FCA axis and the measured physical variables for each region (all sites combined and separately) for each region. Significant correlations ($p \leq 0.05$) are indicated in bold.

HKC	Axis	% var	Year	Total precip	1-mo precip	2-wk precip	Flow	FST	Bed move
All sites	1	30.1	-0.04	-0.09	-0.11	-0.11	-0.60	-0.66	0.24
	2	14.3	-0.20	-0.43	-0.36	-0.18	-0.03	-0.10	-0.16
	3	9.4	-0.16	-0.01	-0.13	-0.07	-0.34	-0.32	-0.33
1D	1		-0.60	-0.42	-0.23	-0.39	-0.53	-0.64	-0.32
	2		-0.39	-0.62	-0.40	-0.17	-0.44	-0.62	-0.62
	3		0.20	0.22	-0.18	0.15	0.05	-0.49	0.05
2D	1		0.32	-0.01	-0.25	-0.16	0.06	-0.05	0.21
	2		0.14	-0.23	-0.26	-0.09	-0.07	0.09	-0.07
	3		-0.35	-0.17	-0.12	-0.15	-0.17	-0.50	-0.31
1P	1		-0.46	-0.43	-0.36	-0.40	-0.56	-0.39	-0.29
	2		-0.54	-0.53	-0.44	-0.47	-0.46	-0.65	-0.22
	3		-0.65	-0.50	-0.24	-0.52	-0.40	-0.54	-0.32
2P	1		0.04	0.07	-0.16	-0.08	-0.07	0.47	0.12
	2		-0.06	-0.44	-0.48	-0.16	-0.34	0.14	-0.38
	3		0.29	0.15	-0.32	0.13	0.04	-0.07	-0.22

BSC	Axis	% var.	Year	Total precip	1-mo precip	Flow	Bed move
All sites	1	38.5	-0.08	0.09	0.39	0.12	0.10
	2	20.3	-0.68	-0.07	-0.02	-0.21	-0.12
	3	13.5	-0.35	-0.77	-0.79	-0.59	-0.72
riffles	1		0.14	-0.18	0.32	-0.04	-0.16
	2		-0.79	-0.04	0.18	-0.25	-0.11
	3		-0.32	-0.71	-0.68	-0.61	-0.68
pools	1		0.07	0.43	0.64	0.43	0.45
	2		-0.54	0.11	0.04	-0.04	0.05
	3		-0.46	-0.82	-0.93	-0.64	-0.77

BFRS	Axis	% var.	Year	Total precip	Flow	Bed move
All sites	1	40.0	-0.51	0.47	0.61	-0.32
	2	16.0	-0.81	0.83	0.20	0.03
	3	10.7	0.09	0.01	0.24	-0.68
BC	1		0.36	-0.21	-0.09	0.00
	2		-0.88	0.83	0.84	-0.79
	3		-0.69	0.83	0.79	-0.41
DC	1		-0.90	0.76	0.76	0.18
	2		-0.83	0.90	0.76	0.39
	3		0.14	0.29	0.17	0.25
DE	1		-0.89	0.89	-0.27	0.85
	2		-0.71	0.61	-0.22	0.70
	3		0.14	0.11	-0.67	0.19
GC	1		-0.93	0.79	0.43	0.64
	2		-0.86	0.95	0.60	0.86
	3		0.74	-0.60	-0.53	-0.75
MC	1		-0.88	0.83	0.33	-0.41
	2		-0.83	0.81	0.33	-0.41
	3		0.69	-0.38	-0.22	0.20

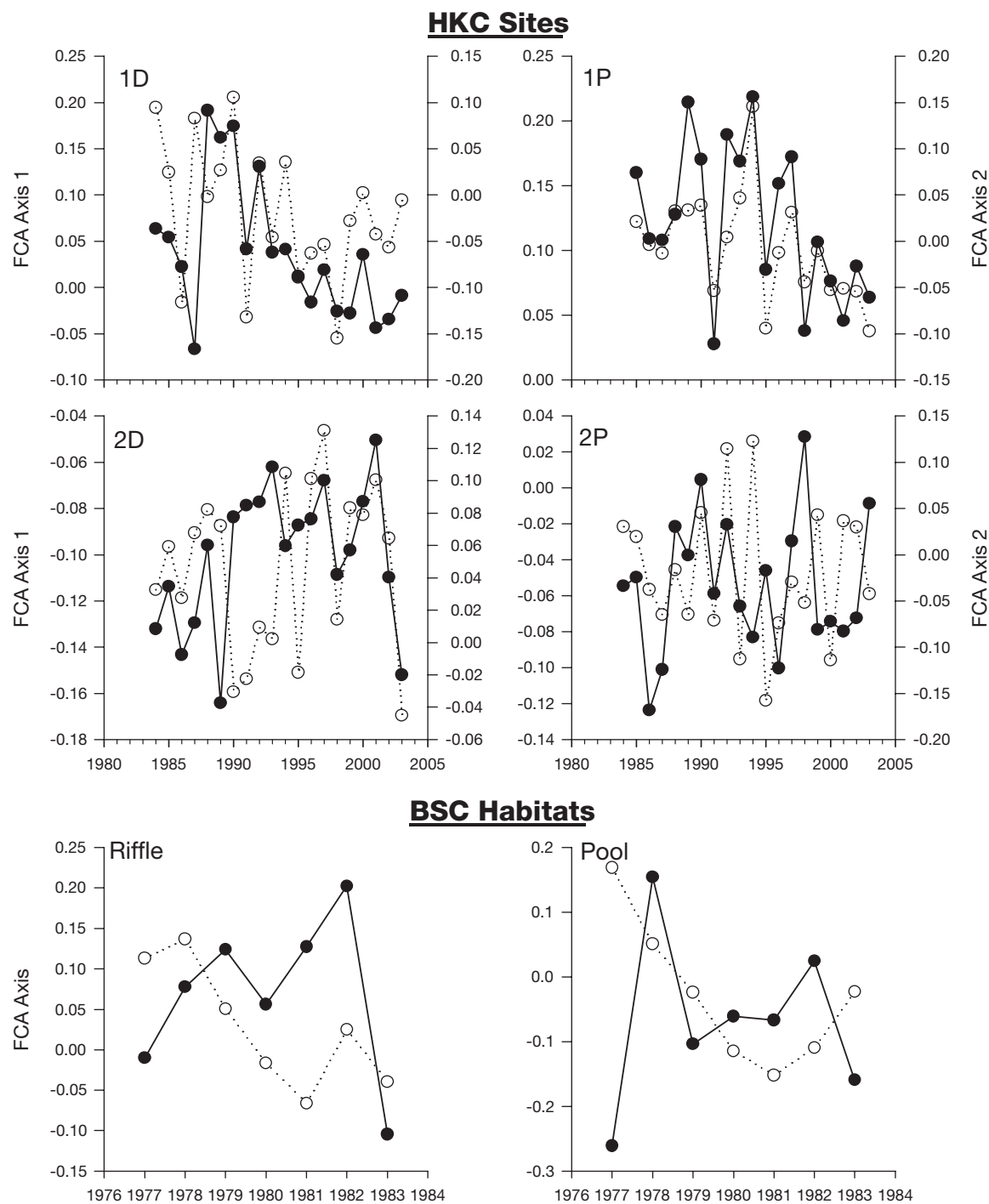


Fig. 4. Temporal variability of the first two FCA axes over time for each site and habitat separately, based on regional FCA (shown in Fig. 3). FCA Axis 1 is represented by closed circles and FCA Axis 2 is represented by open circles. For each HKC site (1D, 1P, 2D, 2P), FCA Axis 2 is plotted on the right on a different scale. FCA Axis 2 is plotted on the same scale as Axis 1 for BSC (riffles and pools) and BFRS (BC, DC, DE, GC, MC) sites.

ences, the regional FCA revealed that the four sites in the HKC catchments were more variable than those in BFRS, as evidenced by the spread of points (variance of HKC sites = 0.007 along axis 2 compared to 0.002 for BFRS sites).

HKC. Differences among sites were the most important factor in the FCA of the HKC region. The FCA of the HKC sites resulted in a 3-axis solution (Fig. 3) that explained 54.7 % of the total inertia of 0.033. Axis 1 represented 30.1 % of the total variance explained,

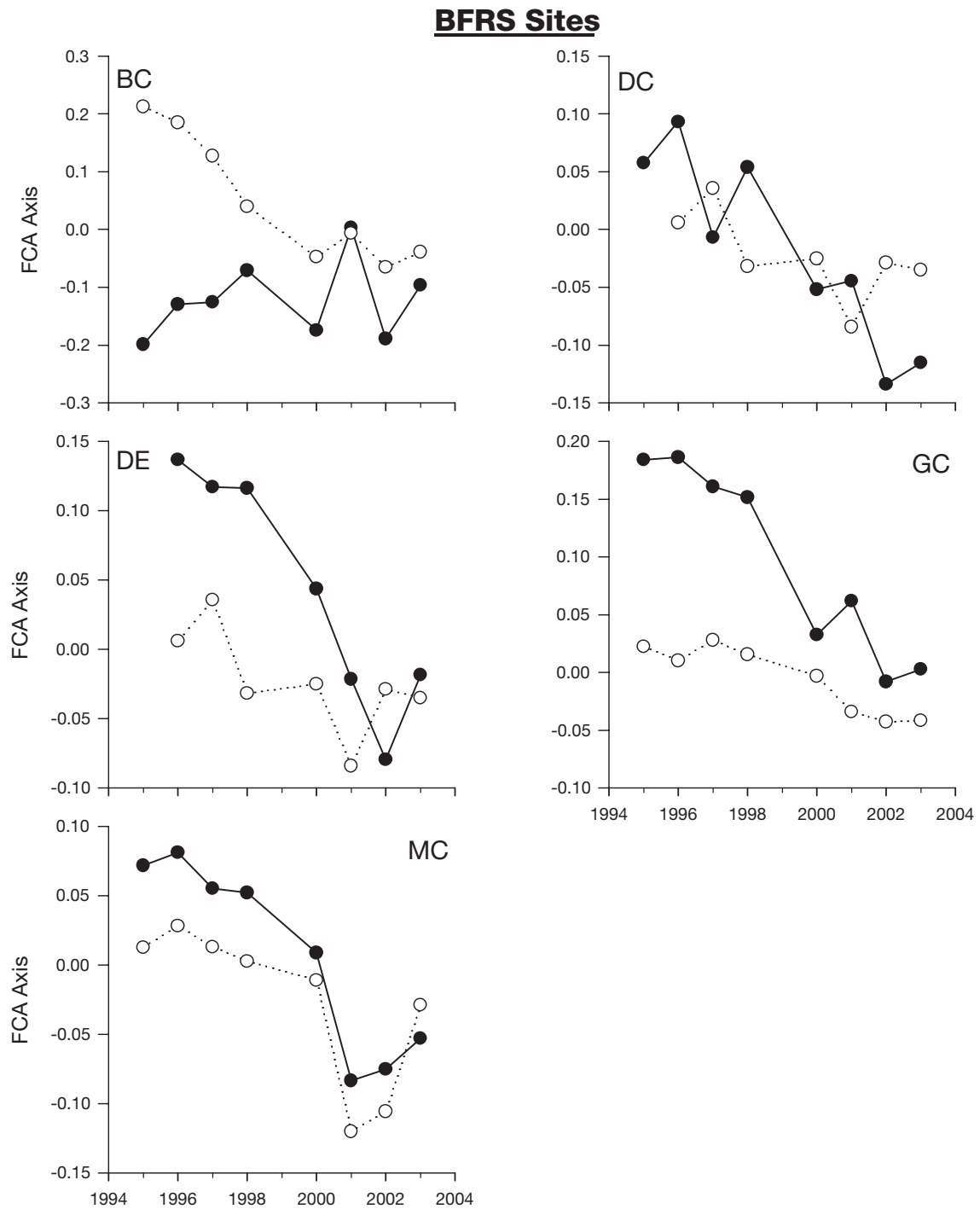


Fig. 4. (continued)

axis 2 represented 14.3 %, and axis 3 represented 9.4 %. When all four sites were analyzed together, axes 1 and 3 were positively correlated ($p < 0.05$) with FST-hemispheres, flow, and bed movement at the time of sampling (Table 4), indicating differences among the sites. These site differences along axes 1 and 3 were confirmed by ANOVA (axis 1: $F = 61.0$, $df = 3$, p

< 0.0001 ; axis 3: $F = 16.6$, $p < 0.0001$), with 1st-order sites (1D, 1P) being distinct from 2nd-order sites (2D, 2P) along axis 1 and sites 1D and 2P being distinct from sites 1P and 2D along axis 3. Axis 2 also revealed site differences ($F = 4.6$, $df = 3$, $p = 0.005$), with site 2D being distinct from the other sites.

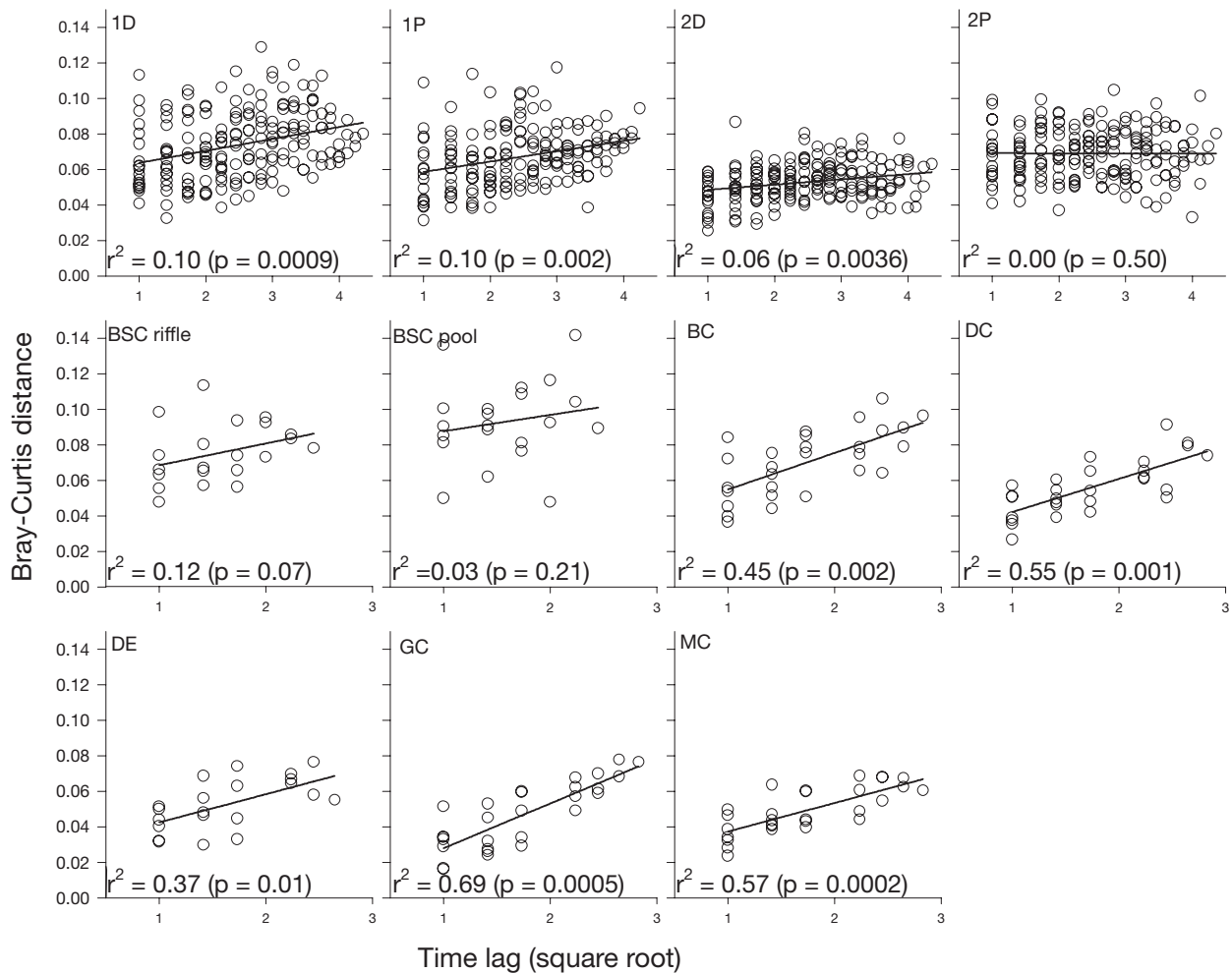


Fig. 5. Time-lag regression analysis results, showing Bray-Curtis distance between each pair of observations versus the square root of the time lag between observations for each site at each region (HKC: 1D, 2D, 1P, 2P; BSC riffles and pools; BFRS: BC, DC, DE, GC, MC). r^2 values are presented based on linear regressions, and significance values are based on 10,000 Monte Carlo simulations.

The relationship of axis 1 with time showed that the biological trait composition of the 1st-order sites (1P, 1D) exhibited directional change; however, the 2nd-order sites did not exhibit significant temporal change (Table 4, Fig. 4). Similarly, correlations between axes and physical and climatic variables were generally stronger for the 1st-order sites than the second-order sites (Table 4). For example, axes 1 and 2 were significantly correlated with flow and total precipitation, respectively, for both sites 1D and 1P; however, at sites 2D and 2P, only axis 2 was correlated with any variable (1-month precipitation) (Table 4). Traits associated with drier years (as determined by positive correlation with axis 2, Table 4) include “very small body size” (a1, < 5 mm), “bi- or multi-voltinism” (b1), “spherical body shape” (k3, e.g. Acarina), “plastron respiration” (o3, e.g. Elmidae), “long-lived adults” (c5, > 365 d),

and “endophytic egg laying” (g4). In contrast, traits associated with wet years include “semi- to mero-voltinism” (b3), “small body size” (a2, 5–10 mm), “filter feeders” (m5), “FPOM feeding” (n1), “cylindrical body shape” (k1), and “aquatic passive dispersal” (drift, h1).

BSC. Temporal trends in biological trait composition based on FCA were largely influenced by the two drought years, particularly in pool habitats. The FCA of the BSC habitats resulted in a 3-axis solution (Fig. 3) that explained 72.3 % of the total inertia of 0.043. Axis 1 represented 38.5 % of the total variance explained, axis 2 represented 20.3 %, and axis 3 represented 13.5 %. Differences between riffle and pool habitats were evident only along axis 1, although this pattern was weak (t-test, $t = 4.6$, $df = 1$, $p = 0.053$). In contrast, axes 1 and 2 revealed different trait composi-

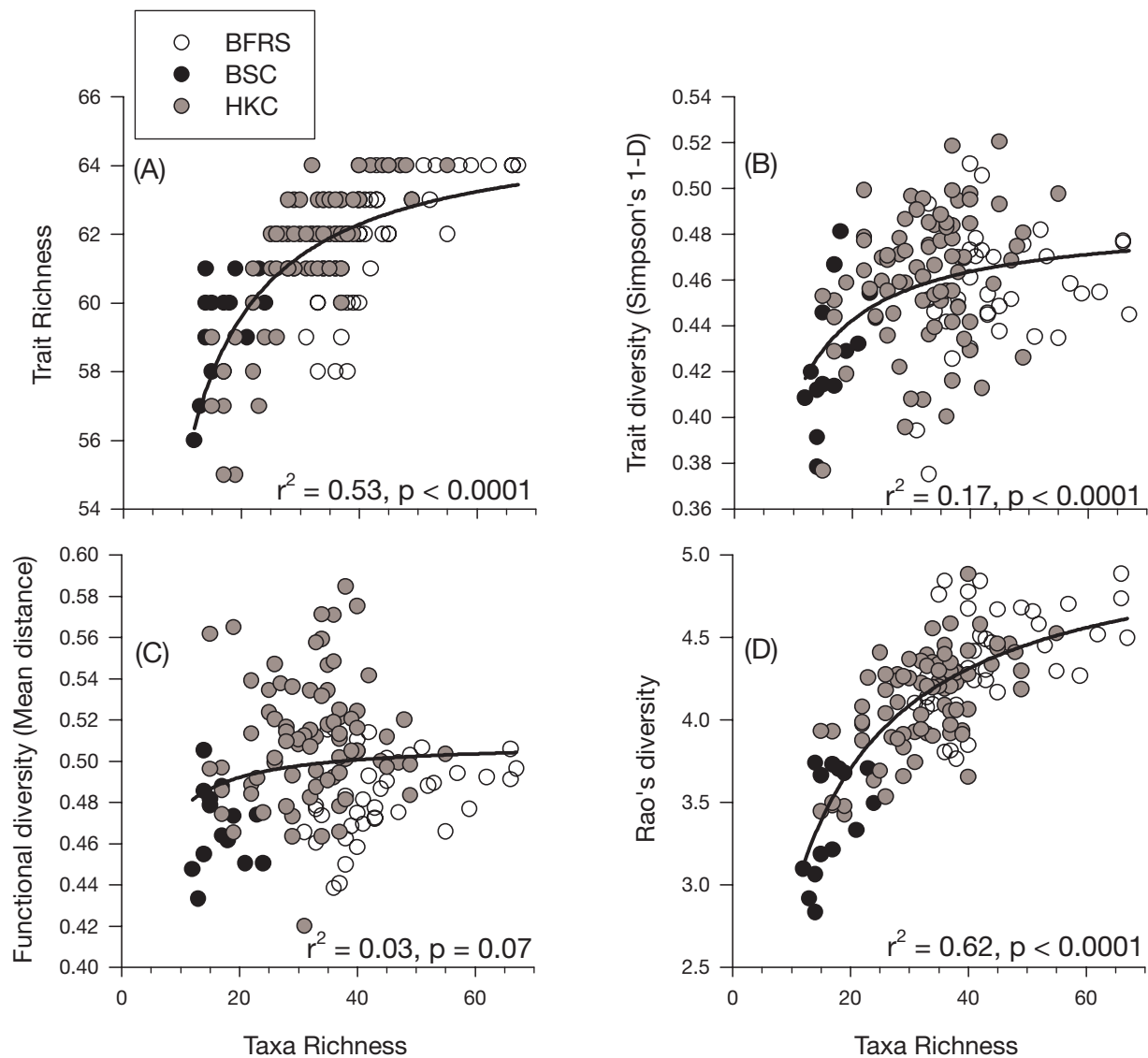


Fig. 6. Relationship between the four measured functional metrics and taxa richness for all regions (open = BFRS; black = BSC; gray = HKC) : (A) Trait richness, TR; (B) Trait diversity, TD; (C) Functional diversity, FD; (D) Rao's quadratic entropy diversity, RD. Non-linear regressions ($y = ax/(b + x)$) are shown.

tion in the initial drought year (1977) for both habitats (Fig. 3, 4), but this pattern was stronger in riffles (Fig. 3). Axis 3 is the only axis significantly correlated with physical and climatic variables when both habitats are combined and for pools alone (Table 4). Traits associated with dry years (as indicated by positive correlation with axis 3) included “moderate body size” (a3, 10–20 mm), “desiccation resistance” (e1), “engulfing predators” (m7), and “aerial respiration” (o4). In contrast, wet-year traits included “semi- to mero-voltinism” (b3), “endobenthic attachment” (i4), “small body size” (a2, 5–10 mm), and “aquatic larvae” (d2).

BFRS. The FCA of the BFRS sites resulted in a 3-axis solution (Fig. 3) that explained 66.7 % of the

total inertia of 0.027. Axis 1 represented 40.0 % of the total variance explained, axis 2 represented 16.0 %, and axis 3 represented 10.7 %. Axes 1 and 2 were significantly correlated with year (Table 4) at all sites, except site BC where only axis 2 varied significantly over time. In addition to significant temporal trends, axis 1 was significantly correlated with total precipitation at all sites, and flow only at site DC (Table 4). Bed movement was significantly correlated with axes only for sites BC, DE, and GC. Both axes 1 and 3 exhibited significant site differences (ANOVA, $df = 4$; axis 1: $F = 8.8$, $p < 0.0001$; axis 2: $F = 10.1$, $p < 0.0001$), with site B2 being distinct along axis 1 and site DE being distinct along axis 3.

Table 5. Spearman rank correlations among trait metrics for all regions combined and for each region separately. Significant correlations ($p < 0.05$) are indicated in bold.

All regions	TD	FD	RD	TR	Taxa diversity	Taxa richness
Trait diversity (TD)	–					
Functional diversity (FD)	0.21	–				
Rao's diversity (RD)	0.46	0.39	–			
Trait richness (TR)	0.21	0.36	0.65	–		
Taxa diversity	0.72	0.02	0.20	0.00	–	
Taxa richness	0.27	0.07	0.72	0.73	0.20	–
HKC	TD	FD	RD	TR	Taxa diversity	Taxa richness
Trait diversity (TD)	–					
Functional diversity (FD)	–0.07	–				
Rao's diversity (RD)	0.37	0.36	–			
Trait richness (TR)	0.07	0.27	0.55	–		
Taxa diversity	0.81	–0.21	0.15	0.12	–	
Taxa richness	0.23	0.10	0.57	0.72	0.27	–
BSC	TD	FD	RD	TR	Taxa diversity	Taxa richness
Trait diversity (TD)	–					
Functional diversity (FD)	0.04	–				
Rao's diversity (RD)	0.23	0.37	–			
Trait richness (TR)	0.22	0.62	0.58	–		
Taxa diversity	0.82	0.24	0.44	0.54	–	
Taxa richness	0.68	0.02	0.52	0.50	0.67	–
BFRS	TD	FD	RD	TR	Taxa diversity	Taxa richness
Trait diversity (TD)	–					
Functional diversity (FD)	0.57	–				
Rao's diversity (RD)	0.59	0.13	–			
Trait richness (TR)	0.19	0.48	–0.52	–		
Taxa diversity	0.38	–0.01	0.94	–0.60	–	
Taxa richness	0.09	0.39	–0.50	0.81	–0.50	–

Traits associated with dry years (as indicated by a negative correlation with axis 2, which exhibited the strongest relationship to total precipitation) include “asexual” (f2) and “parthenogenic” (f3) reproduction, “parasitic” (m8) and “absorbing” (m1) feeding habits, “cutaneous respiration” (o1), and “very large body size” (a5, > 40 mm). Traits associated with wet years include “gill respiration” (o2), “filter feeders” (m5), “no resistant stage” (e3), “short-lived adults” (c2, > 1–10 d), “crawler behavior” (i2), and “univoltinism” (b2).

Long-term trends in variability of biological trait composition

TLR analysis showed positive correlations (directional change) for eight of the 11 sites and habitats, indicating directional change in trait composition (Fig. 5). The strongest relationships were found for the BFRS sites and the first-order sites (1P and 1D) at HKC (Fig. 5). At BSC and site 2P, relationships were weak or insignificant, indicating no temporal trend in biological trait composition (i.e. either stability or stochastic

Table 6. Spearman rank correlations between each trait metric and all climatic and physical variables for all regions combined and for each region separately. Significant correlations ($p < 0.05$) are indicated in bold. (–) signifies no data are available for this variable.

	Year	Total precip	1-mo precip	2-wk precip	Bed move	FST	Flow
<i>All regions</i>							
TD	0.24	–0.07	–0.35	–	–0.11	–	–0.05
FD	0.06	–0.52	–0.43	–	–0.17	–	–0.59
RD	0.47	–0.09	–0.58	–	–0.23	–	–0.4
TR	0.36	–0.35	–0.56	–	–0.20	–	–0.42
<i>HKC</i>							
TD	0.11	0.07	–0.23	0.14	–0.02	0.04	0.14
FD	0.04	–0.22	–0.20	–0.08	–0.01	–0.45	–0.47
RD	0.02	–0.30	–0.40	–0.13	–0.07	–0.20	–0.20
TR	–0.05	–0.49	–0.42	–0.27	–0.06	–0.12	–0.24
<i>BSC</i>							
TD	0.02	–0.24	–0.22	–	–0.25	–	–0.29
FD	–0.64	–0.30	–0.13	–	–0.30	–	–0.33
RD	–0.54	–0.81	–0.76	–	–0.79	–	–0.67
TR	–0.85	–0.62	–0.50	–	–0.65	–	–0.72
<i>BFRS</i>							
TD	0.17	0.06	–	–	0.09	–	–0.35
FD	0.32	–0.22	–	–	–0.07	–	–0.32
RD	0.44	–0.37	–	–	–0.13	–	–0.39
TR	0.80	–0.70	–	–	–0.17	–	–0.16

change with respect to time). For example, distance between years increased with increasing time lags at all BFRS sites, with r^2 values ranging from 0.10 (sites 1P and 1D) to 0.69 (site G2) ($p < 0.05$) (Fig. 5).

Temporal variability of trait diversity and richness

Temporal variability of trait metrics was low at all sites in all regions. For example, in all cases, the CV of all trait metrics was less than or equal to 10 %, with the exception of RD (CV = 32 %). TR was slightly more variable in the BFRS region (mean CV $\pm$ SD = 3.0 % $\pm$ 0.96 %), whereas variability of TD was highest at HKC and BSC (mean CV = 6.0 % $\pm$ 1.1 % at HKC, 6.1 % $\pm$ 1.1 % at BSC). Strong linear trends over time in TR, FD, and RD were observed at both BSC and BFRS (Table 6) that corresponded to the linear increases or decreases in precipitation in each region, respectively. In contrast, only RD exhibited significant temporal trends at HKC (at sites 2D and 1P).

Although the ranges of each of the trait metrics were similar across regions, BSC had significantly lower trait diversity (TD, FD, RD) and trait richness than HKC and BFRS, which had similar values. For example, TR ranged from 54 to 64 trait categories

(mean = 61 ± 2.1) in all three regions, and values at BSC were significantly lower (TR = 59 ± 1.5) than the other two regions (TR = 62 ± 2.0 at HKC and 62 ± 0.9 at BFRS) based on Tukey HSD test following ANOVA ($F = 10.04$, $p < 0.0001$). Similar patterns were found for all three trait diversity metrics as well. TD ranged from 0.38 to 0.52 (mean = 0.46 ± 0.03), and for 68 % of site-years TD ranged between 0.45 and 0.50. FD ranged from 0.42 to 0.58 (mean = 0.50 ± 0.03). RD ranged from 0.17 to 4.17 (mean = 2.41 ± 0.78).

When all three regions were analyzed together, taxa richness was a good predictor of trait metrics. For example, three of the four trait metrics (TD, RD, TR) were positively correlated with taxa richness. TR was positively correlated with taxa richness, and TR leveled off at ~ 50 taxa. There was also a weak, but significant, linear relationship between TD and taxa richness (Fig. 6). Similarly, TD was linearly correlated with taxa diversity (Fig. 6). FD was positively correlated with TD and TR, however FD was not correlated with RD, taxa richness, and taxa diversity. RD was strongly correlated with TR and exhibited an even stronger positive relationship with taxa richness. In contrast, the positive relationship between RD and taxa diversity, although significant ($r_s = 0.20$, $p = 0.05$), was much weaker.

In general, trait diversity and richness metrics decreased with increasing precipitation and flow (Table 6), with higher precipitation resulting in increased flooding and lower taxa richness and trait diversity and richness. For example, FD and TR both decreased with total precipitation, and all four metrics decreased with 1-month precipitation. In contrast, bed movement, 2-week precipitation, and FST showed weak or no relationship with trait metrics.

Trait modalities exhibited high temporal variability reflecting climatic variability. For example, up to 20 % of all trait modalities had a CV > 50 % (Appendix 1). Furthermore, many trait modalities (27–42 %) exhibited significant temporal trends over the course of the study periods at six of the 10 study sites (1P, BC, DC, DE, GC, MC) based on Mann-Kendall trend tests. Of the six sites with a high number of unstable trait modalities (> 16 or > 25 %), five were from BFRS, but only one HKC site (1P) shows this instability. In contrast, at BSC and the other HKC sites (1D, 2D, 2P), only 5–24 % (3–16) of trait modalities exhibited significant temporal trends (Appendix 1). These patterns in temporal variability correspond well with the FCA and TLR analyses, which indicate the greatest temporal change in small streams (1st-order) and in one 2nd-order BFRS stream.

Discussion

Examinations of both spatial (e.g. Lamouroux et al. 2002, 2004, Statzner et al. 2004) and temporal (i.e. seasonal, Snook & Milner 2002, Bêche et al. 2006) variation have shown high stability of biological traits across broad distances and seasons. Despite this high stability, other studies have shown a positive relationship between habitat characteristics and trait composition (e.g. Townsend et al. 1997, Oberdorff et al. 2001, Mérigoux et al. 2001, Mérigoux & Dolédec 2004). However, our study is the first to explicitly examine inter-annual variability of biological traits in terms of overall composition, trends in individual traits, and patterns in trait diversity and taxonomic richness.

Long-term trends and stability of biological trait composition

Prolonged drought and changing temporal patterns in precipitation in the study regions resulted in directional change in biological trait composition. However, biological traits were stable relative to community composition as measured in other studies in this and other datasets (e.g. Bêche 2005, Milner et al. 2006).

This pattern is to be expected because most traits occur in all years and in comparatively constant proportions; however, taxa occurrence and abundance varies from year-to-year, leading to lower stability. Relative stability of biological traits was also shown in seasonal comparisons (Bêche et al. 2006) and comparisons across large spatial scales (Statzner et al. 2004).

Despite the relative stability of biological trait composition in comparison to community composition and abundance, stability analyses (i.e. FCA, TLR) indicated substantial directional change for most sites. For example, for 3 of 4 HKC sites, directional change was observed coinciding with the extended drought period of 1987–1991 (Fig. 5), with the strongest patterns found in the first-order sites (1D, 1P). These first-order streams may be more affected by variations in precipitation and flow caused by drought than larger streams as a result of vegetation encroachment (e.g. *Typhus* sp.). The encroachment of macrophytes resulted in an increase in fine sediment, the narrowing and filling in the channel, and increased near-bed hydraulics; these physical changes resulted in changes in both taxonomic and trait composition (Table 4, Bêche 2005). Two additional disturbances may have contributed to temporal changes in biological trait composition at site 1D (Figs 4, 5). Specifically, the potentially prolonged effects of a toxic sediment spill in February of 1986 (Resh & Jackson 1993) resulted in reduced abundance of *Baetis tricaudatus* (Dodds). In addition, a 1999 wildfire (only affecting site 1D) resulted in a post-fire increase in diversity (from 0.61 ± 0.16 to 0.76 ± 0.06); which may have contributed to the temporal trajectory of community change (Fig. 4).

Communities from BSC and all of the BFRS sites exhibited directional change in biological trait composition (Figs 4, 5), presumably as a result of the linear increase (at BSC) or decrease (at BFRS) in precipitation over the study period (i.e. the first and last year of the study differ dramatically in total precipitation; Fig. 1). For example, extreme drought (1976–1977) may have also affected the biological trait composition of both riffle and pool communities at BSC, resulting in directional change from 1979 to 1983 (axis 2, Fig. 4). However, the pattern of directional change observed at BSC may also be related to an overall increase in precipitation over the study period (Fig. 1). Directional change in biological trait composition was strongest at BFRS, and was strongly correlated with a decrease in total precipitation over the study period (Fig. 1).

Life-history strategies are expected to be related to seasonal flow dynamics (Bêche et al. 2006) because the seasonality of flow is highly predictable in the

mediterranean-climate region (Gasith & Resh 1999). However, extreme droughts are rare and unpredictable in occurrence. In this study, extreme droughts have been shown to cause temporal change in biological-trait composition. Most species lack adaptations to cope with extreme events because these events are so rare (Lytle & Poff 2004), which could lead to unexpected changes in community structure, such as the directional change observed. However, some species have adaptations to these extreme events that result in increased fitness costs during average or wet years, but allow the species to persist (i.e. resistance traits) or quickly recolonize (i.e. resilience traits) (Boulton et al. 1992, Boulton 2003, Lytle & Poff 2004). Some of these adaptations may include diapause, desiccation resistance, or bet-hedging strategies (e.g. asynchronous egg hatching in some Plecoptera, Lytle & Poff 2004; Table 2), some of which were found to vary in response to annual variability of precipitation (data not shown).

Temporal patterns of individual trait modalities

Patterns of the relationship between biological trait composition and environmental variables may be obscured by trade-offs among different adaptations to flow conditions (Resh et al. 1994, Lytle & Poff 2004). Although the multivariate ordinations (FCA) were useful in describing overall temporal variation and aided in the interpretation of stability analysis (i.e. TLR), we found that the FCA resulted in difficult-to-interpret patterns of the relationship between traits and physical variables. Therefore, as suggested by numerous authors (e.g. Townsend et al. 1997, M  rigoux et al. 2001, M  rigoux & Dol  dec 2004), we are relying primarily on the response of each trait to annual variability in precipitation and near-bed hydraulic conditions in the final interpretation of results.

Many of the traits were significantly correlated with precipitation or stream hydraulic variables, which are indicative of the precipitation gradient from dry to wet years (Appendix 2). Trait modalities that differed between wet and dry years included many of the modalities that differed between spring and summer (B  che et al. 2006), which indicates that dry years can result in immediate post wet-season habitats that are similar to dry season habitats (i.e. reduced flow and low shear stress). Thus, communities that were present in dry years had traits that allowed them to survive in low or no flow conditions (e.g. burrower or surface swimmer locomotion), with low available oxygen (aerial respiration), and with increased risk of drying (des-

iccation resistance, diapause, aerial dispersal, long-lived adults). In contrast, communities in wet years were dominated by traits that conferred resilience or resistance to flooding (e.g. Boulton et al. 1992). For example, traits that would enable invertebrates to quickly recolonize (resilience traits) post-flood from other areas (short life spans with > 1 generation per yr) or from upstream habitats (drift dispersal) were present. Traits that confer resistance to flooding, such as flattened body shapes (allowing them to exploit interstitial spaces) and attached eggs were also common. Taxa that were common in wet years also had traits that take advantage of high dissolved oxygen (i.e. cutaneous respiration) from elevated flow.

Using precipitation or bed movement as a proxy for disturbance, relationships between biological traits and disturbance may be obscured in years where there has been ample time between the last flood and sampling date to allow for colonization and establishment of taxa with a range of trait strategies (Townsend et al. 1997). Similarly, we found that 2-week precipitation and FST-hemispheres (with both variables reflect localized habitat conditions immediately prior to and during sampling) were strong predictors of individual trait modalities (data not shown) and biological trait composition (Table 4). These results provide evidence that conditions immediately before sampling were responsible for much of the variation in biological traits. This pattern was apparent at all sites except 2D, where variation in shear stress was low despite higher magnitude of flow fluctuations (compared to the 1st-order sites); this likely results from high embeddedness. However, even with the high correlations between variables indicative of recent flooding, total precipitation is still a strong predictor of biological trait composition, both in terms of individual modalities and overall composition (Table 4). One reason for this strength may be that in years with average precipitation, the amount of 2-week pre-sampling date precipitation may be either very low (e.g. 1999, 2000) or very high (e.g. 1991, 2000), and therefore not reflective of overall, year-to-year climatic patterns (Fig. 1).

Temporal vs. spatial variability of trait diversity and composition

Spatial gradients of trait diversity in streams (e.g. Heino 2005) and other terrestrial ecosystems (e.g. Stevens et al. 2003) have been associated with environmental gradients such as acidity, canopy cover, and substrate characteristics. In contrast, more geographically extensive studies have found high spatial stability in the biological traits (both diversity and composition) of

stream organisms (i.e. macroinvertebrates and fish) across large geographic regions (e.g. Europe; Lamouroux et al. 2002, Statzner et al. 2004).

Although temporal variability of stream communities at the same site might be expected to be lower than spatial variability across sites on the same date (Sandin & Johnson 2000), recent examinations of the HKC dataset used in this study have found the opposite pattern. Contributing to this pattern of higher temporal than spatial variability is the fact that the spatial extent of this study is small. In addition, higher than expected temporal variability in macroinvertebrate communities may be a result of the high inter-annual variability in precipitation in the mediterranean-climate region (e.g. Gasith & Resh 1999) as well as the long-lasting influence of prolonged drought on habitat characteristics. Even though considerable temporal variation in biological traits occurs within a region (Fig. 2), when a broader geographical range is considered, there was more variation between regions than over time in a single region (Fig. 2). For example, in terms of both trait diversity and composition, the BFRS sites were distinct from the other two regions (Fig. 2), which are geographically clustered (Fig. 1). The uniqueness of the BFRS sites may have been a result of biological adaptations to a hydrological regime influenced by snow (15–20 % of the annual precipitation), which created a more stable flow environment (both intra- and inter-annually). For example, increased seasonal and inter-annual stability in flow could result in more long-lived taxa, with both aquatic larvae and adults, and fewer taxa with aerial respiration; these traits were more common at the BFRS sites than in the other two regions.

Temporal variability of trait diversity metrics

Despite high temporal variability in individual trait modalities, trait diversity and richness metrics were relatively stable over time. However, at the same sites, taxonomic turnover was high (Resh et al. 2005). Temporal constancy in trait metrics despite high taxonomic turnover implies high taxonomic redundancy in traits at a given site. Redundancy of traits in taxa is to be expected, particularly in harsh environments where few traits are adaptive (e.g. Poff 1997, Lamouroux et al. 2002, 2004). For example, the observed temporal stability of trait metrics occurred even when the physical environment (e.g. flow, substrate) changed dramatically over time. Furthermore, at HKC, trait metrics were similar between years with extreme drought or flooding conditions even though traits adapted to flooding

(e.g. “flattened body shape”) or drought (e.g. “aerial respiration”) were more common in those years.

Trait diversity metrics have been shown to vary spatially in some studies in response to environmental variables (e.g. Heino 2005); this spatial variability implies that either the range of environmental variability at our sites was too low to produce change in trait metrics or the traits are filtered at spatial scales much larger than that encompassed by the present study. However, other studies using similar trait-coding methods have shown low spatial variation in stream communities for both fish and invertebrates (e.g. Statzner et al. 2004, Lamouroux et al. 2002, 2004). Reasons for low variability in trait metrics include the dominance of a single trait category within a trait (e.g. “sexual reproduction” was the dominant modality for the reproduction trait) and the lack of variability in the occurrence of most traits. Furthermore, most traits occurred in most years, with only the presence of the rarest traits (in terms of the relative abundance of taxa with those traits, e.g. parthenogenesis) varying from year to year.

Although there was low overall variability in trait metrics, we observed strong temporal trends in several metrics at BSC and BFRS, corresponding to the linear increase or decrease in precipitation in each of these regions, respectively. The relationship between trait metrics and abiotic factors, such as flow, was often clear, even if the magnitude of change was small. For example, significant but small changes in biological traits were found between seasons at HKC (Bêche et al. 2006), and among regions in Europe (Statzner et al. 2004). Similarly, when habitat and climatic variables (total rainfall, bed movement, flow) were included in a multiple regression model with region, both region and a habitat or climatic factors were significant explanatory variables for three of the four trait metrics ($p < 0.05$). However, region was the best explanatory variable (or the only) for two metrics (TD and RD), whereas total rainfall or flow was a better explanatory variable for FD and TR. These results imply that abiotic control of trait diversity and richness was strongest for metrics based on presence-absence of taxa, whereas regional differences (which serve as a surrogate for other unmeasured abiotic and or biotic factors) were more important in describing variation in metrics based on relative abundance (TD, RD).

Variation in trait metrics between regions

Trait metrics varied among regions although they were relatively constant at one site over time. For example, BSC had significantly lower values for all trait met-

rics relative to HKC and BFRS sites. BFRS had the highest values for all trait metrics, with HKC being intermediate between the two regions. These results were similar to the pattern found when examining trait composition using FCA, where BFRS sites were found to be distinct from BSC and HKC sites were intermediate in trait structure (Fig. 6). However, despite regional differences in trait metrics, the magnitude of the difference, particularly between BFRS and HKC, was small.

Low taxonomic diversity resulted in low trait diversity and richness. For example, BSC had the lowest values for trait diversity and richness metrics and only 41 taxa occurred at BSC compared to 157 taxa at HKC and 104 taxa at BFRS (including only those taxa present in > 1 year). Low taxa diversity at BSC may be a result of the influence of nearby geothermal springs on the fauna of the study stream (e.g. McElravy et al. 1989). Although fewer total taxa were collected at BFRS than HKC, there was higher taxa richness in an average year at the BFRS sites ($\bar{X} = 44.5 \pm 9.8$) than at the HKC sites ($\bar{X} = 32.6 \pm 8.4$), contributing to the higher trait diversity and richness at BFRS. However, a trait-accumulation curve showed that only 16 taxa were required to represent 90 % of traits; 44 taxa represented 95 % of traits. Similarly, previous studies on these three datasets have shown that rare taxa, in addition to being low in abundance, also have similar trait composition to common taxa (Resh et al. 2005). Thus, the addition of new taxa over time, many of which are rare, would contribute little to trait richness.

Relationship between diversity and richness of traits and taxa

We found high redundancy among the four trait metrics studied despite considerable debate over the appropriate metrics to use to describe trait (or functional) diversity (e.g. Mason et al. 2005, Ricotta 2005). For example, TD, as measured by the average Simpson's diversity of each trait, was positively correlated with each of the other trait metrics ($r_s > 0.20$; Table 5). Some diversity measures account for the presence of a trait with no abundance data (e.g. TR and FD), while other methods take into account the relative abundance (and thus, to some extent the probability that a given individual will have a particular trait) (e.g. TD and RD). A problem with the former measures is that the influence of abundant species in ecosystem processes is ignored; a problem with the latter (e.g. FAD, Mason et al. 2005) is that they can only be used for a single trait. Both Ricotta (2005) and Zoltán (2005) proposed criteria

to evaluate trait diversity metrics, and these separate evaluations came to the conclusion that Rao's quadratic entropy (in this study, RD) was the best method for calculating trait diversity because it can be used with multiple traits and accounts for relative abundance of taxa.

Taxa richness was positively correlated with three of the trait metrics (trait richness and two diversity measures), but not FD (defined by distances). Furthermore, TR and RD exhibited a saturation effect with increasing taxa richness, with no increase in TR after approximately 50 taxa in a given sample. TD exhibited a less clear saturation effect, with increasing TD until approximately 20 taxa; however, there was much more variability in this relationship. Trait richness and FD, which are based solely on the presence or absence of taxa, would be expected to be the most strongly related to number of taxa, whereas metrics which account for the relative or total abundance of organisms in the sample (e.g. TD and RD) would be expected to be less correlated with taxa richness. We found mixed results with regards to this expectation. For example, of the two trait metrics based on presence/absence (TR and FD), only TR was significantly correlated with taxa richness. Similarly, significant correlations were found between both abundance-based trait metrics and taxa richness (FD and RD); this correlation was strongest for RD (Table 5).

The results from our study indicating high functional redundancy among macroinvertebrates are similar to those from other regions and from other ecosystems. For example, Stevens et al. (2003) found a saturating effect of functional group richness, diversity, and dominance with increasing species richness with bat communities from North and South America. Similarly, in a review of observational and experimental studies, Schwartz et al. (2000) found that most studies exhibited a saturating effect between ecosystem function and biodiversity. In contrast, Micheli & Halpern (2005) found a linear relationship between functional richness and diversity and species richness and diversity, indicating low redundancy in coastal marine communities (algae, invertebrates, fish). One potential reason for the difference between high and low redundancy communities is the relative influence of biotic interactions on community assembly and dynamics. For example, in communities with high levels of competition and predation over both short and long time-scales (e.g. coastal marine communities), pressure for resource partitioning may result in an increase in functional diversity (Micheli & Halpern 2005). In contrast, for communities strongly controlled by abi-

otic factors, such as stream invertebrates, high redundancy may be the result of filtering of traits at larger spatial scales (e.g. Poff 1997, Statzner et al. 2004).

In conclusion, we found that the biological traits of macroinvertebrates reflected short-term changes in habitat despite relatively low temporal and spatial variability in the traits themselves. Furthermore, measures of trait diversity were strongly correlated with taxonomic diversity and richness and indicated relatively high redundancy among macroinvertebrates in terms of their traits. The high functional redundancy in streams implies that in these systems, it likely will be easier to conserve functional or trait diversity rather than taxonomic diversity.

Acknowledgements

We thank G. Lamberti and E. McElravy for field assistance, M. Deitch for developing hydrological models, and R. Mazor for ideas, criticism, and feedback on this analysis and manuscript. This research was supported in part by the U.S. EPA's Science to Achieve Research Results Grants (STAR 83243901 awarded to A. Merenlender).

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Submitted: 11. May 2006; accepted: 9 February 2007.

Appendix 1. Coefficient of variation (CV) values (mean/SD) for each trait modality at each site over time. Trait modality codes correspond to those in Table 2.

	1D	1P	2D	2P	BSC pool	BSC riffle	BC	DC	DE	GC	MC
a1	32.4%	23.7%	25.1%	14.3%	28.7%	48.8%	18.5%	19.9%	29.6%	27.9%	11.8%
a2	8.1%	9.0%	9.6%	7.1%	12.2%	14.6%	6.0%	5.6%	7.6%	6.0%	3.2%
a3	14.8%	11.7%	9.6%	12.9%	11.6%	21.6%	6.7%	13.6%	11.8%	13.3%	8.1%
a4	26.1%	24.0%	34.1%	20.3%	33.4%	36.2%	26.9%	18.5%	15.5%	20.5%	19.3%
a5	42.3%	88.7%	26.9%	62.4%	29.6%	33.4%	33.4%	52.4%	27.6%	40.8%	35.4%
b1	42.8%	61.1%	68.7%	35.1%	60.7%	98.8%	35.2%	12.4%	13.7%	12.1%	15.6%
b2	7.7%	11.1%	6.8%	5.4%	8.9%	17.9%	8.3%	6.3%	3.8%	3.4%	4.8%
b3	10.4%	12.4%	7.5%	7.6%	7.0%	12.8%	20.5%	17.8%	12.4%	10.2%	10.9%
c1	26.0%	25.9%	29.7%	14.2%	18.7%	31.6%	20.3%	17.3%	9.9%	9.4%	18.0%
c2	8.3%	10.6%	12.7%	8.2%	13.1%	20.1%	9.7%	12.3%	11.9%	9.3%	7.9%
c3	27.0%	19.2%	17.3%	21.0%	17.6%	31.7%	8.8%	11.7%	9.9%	9.9%	5.3%
c4	19.1%	12.0%	10.2%	11.5%	37.1%	32.0%	23.7%	30.4%	30.1%	41.0%	19.1%
c5	39.8%	35.1%	19.4%	25.8%	27.5%	29.1%	32.5%	24.9%	31.6%	39.8%	34.2%
d1	12.4%	5.1%	7.3%	2.9%	9.0%	7.2%	5.1%	4.1%	2.0%	2.6%	3.2%
d2	6.5%	4.7%	5.3%	4.4%	7.0%	6.3%	2.6%	2.2%	3.1%	1.8%	1.7%
d3	32.8%	21.1%	16.5%	26.8%	26.5%	28.3%	33.1%	28.7%	35.2%	56.9%	18.7%
e1	61.3%	76.0%	44.8%	66.2%	51.3%	78.5%	22.7%	49.7%	28.3%	47.0%	72.1%
e2	14.5%	13.0%	13.0%	9.5%	12.8%	19.8%	17.9%	10.2%	16.1%	16.3%	10.7%
e3	3.7%	5.5%	4.5%	3.9%	5.0%	5.2%	5.1%	5.2%	2.7%	4.8%	4.8%
f1	1.0%	0.9%	1.1%	0.9%	1.2%	1.2%	1.6%	1.3%	1.0%	1.0%	1.6%
f2	74.4%	67.8%	28.2%	40.5%	42.1%	37.5%	68.1%	74.1%	63.3%	79.0%	93.2%
f3	164.9%	164.2%	82.2%	189.4%	0.0%	0.0%	136.3%	109.4%	55.7%	95.4%	79.4%
g1	22.5%	22.5%	21.2%	18.0%	30.2%	30.1%	18.7%	13.6%	8.8%	10.3%	17.1%
g2	5.4%	5.3%	5.9%	5.0%	5.6%	5.8%	6.3%	3.4%	4.9%	1.8%	2.6%
g3	20.7%	30.2%	28.4%	35.6%	25.3%	52.3%	19.8%	28.4%	24.5%	24.5%	40.3%
g4	40.2%	32.5%	40.7%	21.7%	82.5%	116.6%	72.4%	60.7%	54.2%	91.9%	79.7%
g5	58.7%	36.7%	18.5%	66.3%	0.0%	0.0%	51.1%	72.2%	83.1%	120.0%	52.4%
g6	252.1%	0.0%	0.0%	447.2%	0.0%	0.0%	0.0%	282.8%	135.3%	208.1%	197.8%
h1	10.9%	11.6%	11.4%	7.9%	7.1%	8.5%	6.7%	5.7%	6.5%	2.4%	6.5%
h2	9.3%	9.0%	7.7%	5.7%	4.0%	13.4%	6.0%	6.6%	4.8%	6.4%	8.5%
h3	20.7%	19.8%	24.5%	17.2%	12.4%	25.8%	19.0%	15.0%	14.4%	9.2%	11.4%
h4	10.1%	7.9%	10.6%	6.5%	9.5%	11.4%	13.7%	8.0%	9.7%	4.9%	8.1%
i1	14.3%	16.8%	10.7%	13.3%	12.2%	19.6%	16.7%	22.1%	25.6%	25.4%	16.5%
i2	8.3%	6.9%	4.9%	5.5%	7.7%	8.0%	7.9%	7.8%	7.1%	5.7%	7.1%
i3	17.5%	17.2%	12.0%	13.3%	32.5%	24.0%	19.4%	25.5%	23.4%	31.7%	26.2%
i4	23.6%	30.2%	15.5%	18.6%	30.4%	40.4%	20.3%	17.8%	10.3%	9.6%	21.1%
i5	17.2%	22.5%	30.4%	9.6%	26.8%	47.3%	21.4%	35.5%	21.9%	20.7%	22.1%
i6	108.3%	129.9%	183.1%	50.2%	264.6%	264.6%	282.8%	0.0%	171.5%	282.8%	282.8%
j1	8.3%	10.6%	8.0%	9.4%	27.5%	33.7%	11.2%	11.9%	6.1%	5.0%	4.6%
j2	5.7%	7.7%	5.9%	5.2%	14.7%	13.3%	6.2%	5.2%	2.2%	2.2%	2.1%
k1	21.5%	23.4%	21.2%	20.3%	43.0%	50.1%	13.0%	4.8%	10.8%	4.2%	7.7%
k2	8.2%	7.8%	9.4%	6.1%	5.2%	7.6%	6.9%	3.9%	5.0%	3.0%	2.9%
k3	42.5%	46.8%	33.9%	24.1%	60.5%	58.3%	35.3%	40.3%	35.7%	54.4%	32.9%
l1	10.7%	6.5%	6.3%	6.5%	10.2%	15.3%	10.9%	5.2%	5.4%	7.5%	6.7%
l2	24.8%	15.4%	8.8%	13.5%	26.6%	16.5%	22.7%	13.9%	12.6%	15.4%	11.3%
l3	90.7%	91.2%	106.3%	35.2%	56.1%	70.0%	49.3%	37.3%	50.9%	28.7%	43.8%
m1	58.1%	127.3%	28.3%	76.0%	30.1%	36.7%	60.9%	54.2%	31.7%	51.2%	36.7%
m2	11.6%	9.7%	9.2%	10.1%	12.4%	9.0%	12.1%	10.9%	10.7%	7.6%	9.8%
m3	17.8%	18.8%	14.8%	15.6%	21.3%	23.7%	16.1%	7.4%	12.5%	9.7%	7.4%
m4	13.8%	9.3%	15.7%	11.0%	8.9%	6.3%	9.2%	10.4%	17.8%	6.3%	9.1%
m5	36.4%	28.9%	36.2%	18.7%	39.5%	71.8%	26.8%	38.2%	21.1%	29.1%	31.4%
m6	45.0%	64.6%	38.1%	45.7%	126.1%	129.1%	102.5%	86.1%	83.0%	136.7%	150.1%
m7	12.8%	18.5%	18.1%	13.2%	12.4%	22.8%	15.1%	12.6%	6.9%	8.5%	4.2%
m8	62.6%	71.7%	62.4%	29.7%	83.3%	121.5%	93.6%	113.9%	102.2%	93.6%	131.9%
n1	14.4%	12.6%	11.4%	8.3%	8.9%	12.6%	9.8%	3.4%	5.1%	2.2%	7.1%
n2	20.2%	12.8%	20.9%	15.4%	12.0%	20.5%	14.0%	10.6%	9.2%	8.0%	6.0%
n3	13.7%	10.8%	10.3%	6.0%	7.9%	3.6%	9.0%	5.5%	8.9%	6.2%	5.5%
n4	13.7%	12.5%	20.6%	12.1%	19.2%	18.8%	17.7%	19.1%	10.8%	15.7%	20.3%
n5	24.3%	27.7%	50.4%	17.9%	33.8%	45.9%	18.3%	22.0%	22.1%	16.3%	15.8%
n6	15.8%	15.2%	9.6%	7.6%	20.9%	12.7%	17.5%	15.2%	19.5%	18.4%	29.3%
n7	23.1%	27.9%	27.2%	18.2%	22.9%	33.6%	25.5%	16.3%	8.3%	9.1%	7.1%
n8	167.0%	198.4%	135.7%	89.8%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
o1	9.6%	10.9%	10.2%	5.7%	6.9%	11.1%	13.0%	6.2%	5.4%	3.0%	8.0%
o2	11.4%	13.4%	14.9%	9.9%	12.1%	8.8%	9.5%	7.6%	8.7%	10.8%	14.8%
o3	123.9%	67.6%	48.3%	45.9%	50.4%	111.7%	42.3%	33.4%	24.8%	32.3%	19.4%
o4	23.7%	24.0%	31.2%	14.0%	95.9%	80.9%	42.8%	21.0%	40.0%	43.8%	23.9%

Appendix 2. Mann-Kendall trend test results (τ value) for each study site. Bold values indicate significance at $p \leq 0.05$. Trait modality codes correspond to those in Table 2.

	1D	1P	2D	2P	BSC pool	BSC riffle	BC	DC	DE	GC	MC
a1	-0.48	0.01	-0.16	-0.24	-0.43	-0.05	-0.36	0.36	0.52	0.86	0.79
a2	0.32	0.36	0.08	0.25	0.24	0.62	0.29	-0.14	-0.43	-0.07	-0.71
a3	0.44	-0.38	0.01	0.04	-0.05	-0.24	-0.57	-0.57	-0.62	-0.79	-0.50
a4	0.05	-0.16	-0.03	-0.07	0.05	-0.14	0.29	0.14	-0.24	-0.79	0.50
a5	0.07	0.26	0.36	0.24	-0.33	-0.14	0.79	0.36	0.43	0.21	0.21
b1	0.14	-0.05	0.15	0.03	-0.43	-0.52	-0.36	-0.50	-0.43	-0.50	0.14
b2	-0.21	0.05	-0.16	0.27	0.33	-0.05	-0.57	-0.86	-0.24	-0.43	-0.86
b3	0.04	-0.09	-0.14	-0.26	0.14	0.14	0.64	0.86	0.62	0.86	0.64
c1	0.40	0.08	-0.24	0.20	0.52	0.43	0.43	0.07	-0.71	0.00	-0.86
c2	0.24	0.24	-0.26	-0.09	-0.14	0.05	0.21	-0.57	-0.62	-0.86	-0.29
c3	0.52	-0.03	-0.05	0.14	-0.05	-0.05	-0.07	-0.43	-0.43	-0.50	-0.57
c4	-0.45	-0.26	0.22	-0.17	-0.14	-0.24	-0.36	0.71	0.90	0.79	0.43
c5	-0.17	-0.24	0.19	-0.17	-0.33	-0.33	-0.07	0.14	0.62	0.50	0.57
d1	0.43	0.67	0.16	0.12	0.14	-0.24	0.50	-0.43	-0.33	-0.71	-0.79
d2	-0.12	-0.32	-0.26	-0.12	0.05	0.52	0.43	-0.14	-0.71	-0.50	0.29
d3	-0.42	-0.09	0.33	-0.16	-0.33	-0.71	-0.57	0.57	0.71	0.71	0.21
e1	-0.29	-0.33	0.11	0.13	0.05	-0.24	-0.29	0.64	0.14	0.50	0.79
e2	0.17	0.22	0.05	0.29	-0.24	0.14	0.64	0.43	0.43	1.00	0.50
e3	-0.02	-0.10	-0.13	-0.31	0.05	-0.43	-0.29	-0.71	-0.81	-0.79	-0.79
f1	-0.12	-0.10	-0.41	-0.12	0.05	0.14	-0.43	-0.57	-0.71	-0.71	-0.50
f2	0.02	-0.04	0.20	0.03	-0.05	-0.14	0.14	0.57	0.52	0.64	0.21
f3	0.15	0.07	0.30	0.04	0.00	0.00	0.72	0.73	0.62	0.84	0.69
g1	0.26	0.64	0.15	0.37	0.05	-0.14	-0.57	-0.57	-0.33	-0.79	-0.71
g2	0.17	0.10	-0.09	-0.29	0.24	0.43	0.50	-0.29	-0.43	-0.36	-0.14
g3	-0.24	-0.16	-0.24	0.08	-0.14	-0.14	-0.07	0.43	0.14	0.79	0.57
g4	-0.04	-0.39	0.21	0.17	-0.78	-0.72	-0.71	0.29	0.05	0.57	0.62
g5	0.02	-0.11	0.23	-0.24	0.00	0.00	0.50	0.62	0.71	0.79	0.43
g6	-0.10	0.00	-0.12	0.00	0.00	0.00	0.00	0.36	0.51	0.58	0.37
h1	0.13	0.50	0.08	-0.13	-0.33	-0.24	0.43	0.29	0.52	-0.21	0.07
h2	0.12	0.30	0.02	0.01	0.14	-0.52	0.21	0.07	0.24	0.50	0.29
h3	-0.14	-0.15	0.00	0.05	0.71	0.14	0.57	0.14	0.05	0.36	-0.07
h4	-0.13	-0.54	-0.26	0.04	-0.05	0.33	-0.64	-0.50	-0.52	-0.71	-0.57
i1	-0.59	-0.20	-0.04	0.14	0.33	0.05	-0.36	0.64	0.43	0.79	0.71
i2	0.27	-0.12	-0.14	-0.02	0.14	-0.24	-0.50	-0.71	-0.62	-0.86	-0.79
i3	0.21	-0.26	0.16	0.08	-0.43	0.05	0.79	0.71	0.90	0.71	0.64
i4	0.17	0.27	0.21	-0.31	-0.14	-0.05	0.71	0.14	0.33	0.14	0.29
i5	-0.06	0.60	-0.29	-0.07	-0.14	-0.43	-0.07	-0.29	-0.52	-0.50	-0.50
i6	-0.03	-0.58	0.02	-0.04	0.53	-0.53	0.50	0.00	0.07	0.07	0.21
j1	-0.45	0.04	0.13	-0.09	0.52	0.05	0.00	0.86	0.24	0.36	-0.07
j2	0.45	-0.04	-0.13	0.09	-0.52	-0.05	0.00	-0.86	-0.24	-0.36	0.07
k1	0.16	-0.35	0.14	-0.04	0.43	0.43	-0.21	-0.21	0.14	-0.21	-0.57
k2	0.14	0.22	-0.14	0.03	0.05	-0.14	0.50	-0.36	-0.52	-0.64	-0.07
k3	-0.27	-0.10	0.16	-0.06	-0.71	-0.43	-0.57	0.43	0.81	0.50	0.50
l1	0.49	0.35	-0.14	0.29	0.33	0.24	0.64	-0.64	-0.43	-0.79	-0.57
l2	-0.54	-0.64	0.26	-0.41	-0.52	-0.52	-0.86	0.50	0.33	0.21	0.57
l3	0.23	0.26	0.18	0.14	-0.33	-0.05	0.29	0.21	0.14	0.93	0.36
m1	0.08	0.29	0.35	0.18	-0.24	-0.24	0.21	0.43	0.33	0.36	0.21
m2	0.36	-0.24	0.55	-0.11	0.52	0.24	0.57	0.21	0.52	0.29	0.21
m3	0.04	0.15	0.17	-0.11	-0.05	0.43	0.29	-0.14	-0.33	0.21	-0.36
m4	0.13	0.11	-0.15	-0.19	0.05	0.05	0.29	-0.50	-0.62	0.07	-0.21
m5	0.16	0.68	0.05	0.12	-0.05	0.24	-0.07	-0.07	-0.24	-0.64	-0.21
m6	-0.32	-0.06	-0.09	0.13	-0.10	0.06	-0.76	0.11	0.10	0.40	0.56
m7	-0.06	-0.47	-0.28	0.09	-0.24	-0.05	-0.71	0.07	0.05	-0.36	0.86
m8	0.04	0.30	-0.04	-0.19	-0.29	-0.41	0.55	0.87	0.51	0.62	0.79
n1	0.11	0.47	0.27	0.02	-0.05	0.43	0.57	0.14	-0.52	-0.14	-0.50
n2	0.25	0.33	0.25	-0.05	0.14	0.05	0.14	-0.29	-0.05	0.07	-0.57
n3	0.16	0.36	-0.15	-0.15	-0.05	0.24	0.50	-0.21	-0.05	0.29	-0.14
n4	0.03	0.13	0.04	-0.13	0.62	0.24	-0.21	0.21	0.24	-0.36	-0.07
n5	0.34	-0.63	0.17	0.35	-0.43	-0.33	0.43	0.43	0.90	0.57	0.64
n6	-0.18	-0.17	-0.11	-0.24	-0.24	0.14	0.43	0.57	0.43	0.43	0.64
n7	-0.22	-0.36	-0.35	0.12	-0.05	-0.33	-0.79	0.07	0.24	-0.14	0.57
n8	0.21	0.19	0.14	0.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00
o1	0.12	0.33	-0.19	-0.22	-0.05	-0.14	0.71	0.36	0.24	0.71	0.43
o2	0.20	0.01	-0.07	0.02	0.14	0.43	-0.14	-0.07	-0.62	-0.64	-0.21
o3	0.19	0.12	0.32	-0.07	-0.14	-0.10	-0.21	-0.21	0.52	0.36	0.43
o4	-0.18	-0.38	0.38	0.23	-0.20	-0.14	-0.93	-0.57	0.52	0.50	-0.29